Optical Circuit Switching for AI and Hyperscale Datacenters: Benefits, Technologies, Current and Future Deployments

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Executive Summary

Optical Circuit Switching (OCS) has emerged as a critical technology for next-generation Artificial Intelligence (AI) and hyperscale data-center networks. Traditional Electrical Packet-Switch (EPS) fabrics increasingly struggle with congestion, power consumption, and scalability constraints as modern Artificial Intelligence/Machine Learning (AI/ML) workloads demand higher bandwidth, predictable latency, and large-scale synchronization. OCS overcomes these challenges by providing fully transparent, photonic connections without Optical-Electrical-Optical (O-E-O) conversion, enabling ultra-low latency, zero buffering, and protocol-agnostic operation. These characteristics significantly reduce network power and jitter while improving throughput and efficiency. This Open Compute Project (OCP) white paper surveys major OCS technologies, including robotic mechanisms, Micro-Electro-Mechanical-System (MEMS) beam steering, liquid-crystal devices, piezo-actuated systems, and silicon-photonics switches, comparing trade-offs in radix, insertion loss, switching time, and scalability. It further examines OCS applications in current and future deployments.

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1. Introduction

This white paper adheres to the Open Compute Project (OCP) core tenets of openness, efficiency, impact, scale, and sustainability, as defined by the OCP Foundation. While this document is a technical white paper rather than a hardware or software specification, it is intentionally structured to align with OCP's philosophy of open collaboration, transparent knowledge sharing, and at-scale innovation.

2. Compliance with OCP Tenets

The following subsections describe how the content and intent of this white paper support each OCP tenet.

2.1. Openness

This white paper embodies the principle of openness by providing a vendor-neutral, technology-agnostic survey of Optical Circuit Switching (OCS) architectures, implementations, and deployment models. It draws from publicly available research, industry deployments, and peer-reviewed publications to present a transparent view of the state of the art in OCS for AI and hyperscale data centers.

Rather than promoting a proprietary solution, the document compares multiple OCS technologies—including robotic switching, MEMS, liquid crystal, silicon photonics, metasurfaces, and hybrid approaches—highlighting trade-offs, limitations, and areas for future innovation. By openly documenting lessons learned from large-scale deployments and academic research, this white paper lowers the barrier for adoption, experimentation, and further development by the broader OCP community.

In alignment with OCP's open collaboration model, the content is intended to stimulate discussion, inform architectural decisions, and encourage cross-project engagement among network operators, system architects, silicon vendors, and researchers.

2.2. Efficiency

Efficiency is a central theme of this white paper. Optical Circuit Switching is presented as a fundamentally efficient alternative to traditional Electrical Packet Switching (EPS) for selected data-center and AI networking use cases.

The paper documents how OCS eliminates repeated optical-electrical-optical (O-E-O) conversions, packet buffering, and store-and-forward processing—leading to substantial reductions in power consumption, latency, and jitter. Multiple examples demonstrate how OCS improves network efficiency through:

- Reduced network power draw and operational expenditure,
- Lower end-to-end latency and near-zero jitter,
- Improved bandwidth utilization through topology matching and dynamic reconfiguration,
- Reduced capital expenditure by simplifying network layers and extending system lifetimes.

By quantifying and explaining these efficiency gains in practical deployments, the white paper aligns strongly with OCP's emphasis on measurable improvements valued by end users.

2.3. Impact

The white paper highlights the industry-level impact of Optical Circuit Switching by documenting real-world deployments and forward-looking architectures that are reshaping hyperscale and AI infrastructure. OCS is shown to be a key enabler of:

- Large-scale AI training clusters with predictable performance,
- Physically reconfigurable network topologies that adapt to workload demands,
- New resiliency models based on physical-layer reconfiguration,
- Emerging applications such as quantum networking and optical resource disaggregation.

By aggregating insights from hyperscalers, academia, and system vendors, the document demonstrates how OCS is moving from a niche technology to a foundational building block for next-generation data centers. This impact aligns directly with OCP's mission to accelerate industry transformation through shared innovation.

2.4. Scale

Scalability is addressed throughout the white paper by examining OCS deployments and architectures designed for hyperscale environments. The document discusses how OCS systems can support large port counts, incremental expansion, and flexible topology evolution without the architectural penalties typically associated with scaling electrical packet-switched fabrics. Examples include:

- Spine-layer replacement in large data centers,
- Incremental growth models that avoid disruptive network upgrades.

The white paper also emphasizes the importance of control-plane orchestration, automation, and software integration to ensure that OCS-based solutions can be deployed, managed, and scaled efficiently. This focus on practical scalability aligns with OCP's requirement that innovations be deployable at meaningful scale.

2.5. Sustainability

Sustainability is a core consideration of this white paper and a required OCP tenet. OCS is presented as a technology that directly supports sustainability goals through significant reductions in network power consumption and improved utilization of physical infrastructure. Key sustainability contributions discussed include:

- Lower energy consumption through elimination of power-hungry electronic switching elements,
- Extended infrastructure lifetimes enabled by protocol- and rate-agnostic optical paths,

By enabling higher compute efficiency per watt and supporting the continued growth of AI infrastructure within environmental constraints, Optical Circuit Switching aligns strongly with OCP's commitment to responsible, sustainable innovation.

3. OCS Introduction

Modern data centers and high-performance computing environments are experiencing rapid growth driven by AI/ML workloads. In this context, traditional electrical packet-switching (EPS) architectures face bandwidth bottlenecks, increased latency, and significant power consumption challenges at large scale. Because of these limitations, Optical Circuit Switching (OCS) technology is gaining increasing attention and momentum.

OCS devices establish direct photonic connections between endpoints without requiring optical-electrical-optical (OEO) conversion. Operating purely in the optical domain is one of the key reasons OCS can substantially reduce power consumption compared to EPS. OCS is bufferless and both protocol- and line-rate-agnostic, enabling flexible, ultra-low-latency, high-bandwidth interconnects.

Since OCS creates direct optical connections in a fully transparent mode—agnostic to protocol and data rate—it operates in a jitter-free manner, eliminating packet-buffer-related congestion issues in the network. Additionally, OCS enables new network architectures that support enhanced network partitioning and failover paradigms through dynamic topology reconfiguration.

Various underlying technologies can be used to build an OCS system, each with distinct characteristics. These characteristics include port count (radix), switching/reconfiguration time, signal loss, latching versus non-latching operation, cost, complexity, and reliability, along with additional features such as integrated Optical Power Monitoring (OPM), Semiconductor Optical Amplifiers (SOAs), and Variable Optical Attenuators (VOAs). Examples of OCS implementation technologies include robotic mechanisms, MEMS, liquid crystal, piezoelectric systems, silicon photonics, and hybrid combinations of these. These technologies will be discussed in greater detail in the next section.

The following summarizes the key benefits of using OCS, along with some limitations compared to EPS solutions.

Benefits Of OCS

- **Ultra-low latency and congestion-free operation** enabled by fully optical signal paths. Unlike conventional EPS networks, OCS introduces no packet buffers that can congest or add queuing delays.
- **High throughput and scalability**, supporting data-rate-agnostic operation and large port counts suitable for next-generation network speeds.
- **Energy and cost efficiency** due to the elimination of OEO conversions, resulting in substantially lower power usage than electrical switching. Large-scale deployments have demonstrated up to a 41% reduction in power consumption and a 30% reduction in capex [1].
- **Future compatibility**, as OCS is inherently both protocol-agnostic and data-rate-agnostic.
- **Enhanced flexibility and resiliency**, enabling dynamic topology reconfiguration, improved network partitioning, better resource utilization, and robust failover capabilities.

Limitations Of OCS

- **Circuit switching (one-to-one) vs. packet switching (one-to-many):**
  OCS does not provide packet-by-packet forwarding capabilities, whereas EPS fabrics can naturally fan out traffic and make per-packet forwarding and load-balancing decisions.
- **No differential packet handling, inspection, or packet-level telemetry:**
  Because OCS operates purely in the optical domain and lacks packet-processing logic, it cannot perform functions such as QoS enforcement, packet filtering, traffic inspection, buffering, or per-packet analytics.
- **Insertion-loss constraints:**
  Optical insertion loss limits practical OCS hop counts to approximately one or two without the use of high-power lasers and/or optical amplification. In contrast, EPS fabrics can scale across many hops due to electronic regeneration at each switching stage.

4. General OCS Technologies

Robotic Mechanisms:

These systems function like automated fiber patch panels, where motorized components physically move and connect fiber cables to create new optical paths. Connections are mechanical and latching, remaining intact without power once established. They support very large-scale, non-blocking any-to-any connectivity through path-planning algorithms designed to prevent fiber tangling. The primary drawback of robotic-mechanism OCS is their relatively high reconfiguration time, making them less suitable for highly dynamic environments.

MEMS Mirror (Micro-Electro-Mechanical Systems):

MEMS-based OCS solutions use tiny tiltable mirrors—arranged in 2D or 3D arrays—to steer light beams in free space from one fiber port to another, acting as microscopic optical traffic directors. These devices redirect collimated (parallel, non-diverging) beams bidirectionally between ports. MEMS switches can support large port counts, offer reasonably low optical loss, and operate across wide optical bandwidths. However, MEMS designs can be sensitive to vibration and require precise alignment for stable long-term performance.

Liquid Crystal:

Liquid-crystal–based OCS systems electronically control light paths without moving parts by manipulating optical polarization. By applying voltage across liquid crystal cells, the system can steer, attenuate, or block light. These devices have low driving-voltage requirements and benefit from solid-state operation, but they are polarization-dependent and therefore require polarization-diversity components to handle real-world optical signals effectively.

Piezoelectric:

Piezoelectric mechanisms create precise physical motions when voltage is applied to deform specialized materials, enabling accurate alignment of fibers, mirrors, or couplers—similar to autofocus systems in cameras. Piezoelectric actuators are commonly used in high-precision beam-steering or fiber-alignment OCS designs. A key advantage is excellent precision and repeatability when combined with feedback-based position control. The main limitation is mechanical complexity: systems may require many actuators and precision-assembled components.

Silicon Photonics (Waveguide-based):
Silicon-photonic OCS routes light through submicron on-chip waveguides, steering optical paths using thermo-optic or electro-optic effects—similar to how integrated circuits route electrons, but with photons. These systems employ photonic building blocks such as Mach-Zehnder interferometers, ring resonators, and phase shifters on CMOS-compatible platforms. Major benefits include extremely fast reconfiguration times and the potential for low-cost mass production. A significant challenge is higher insertion loss and crosstalk as port counts increase, although integrating Semiconductor Optical Amplifiers (SOAs) can help compensate for these losses.

Metasurface:
Metasurface OCS steers collimated beams in free space, without mechanical motion, using electrically tunable sub-wavelength pixels that control optical phase. Because the pixels are sub-wavelength, steering can occur over large angles (e.g. ±80°), enabling high port density in a compact volume. Pixels can be polarization-invariant, eliminating the need for polarization-diversity optics, and offer fast switching speeds. As with other free-space approaches, insertion loss does not scale strongly with radix. The main limitations are moderate insertion loss and some wavelength dependence due to pixel dispersion.

Hybrid/Combinations:
Hybrid OCS technologies combine multiple mechanisms—for example, silicon photonics integrated with MEMS micro-actuators—to take advantage of the strengths of each approach. Examples include silicon-photonic switches with vertical adiabatic couplers actuated by MEMS, or systems combining space-switching and wavelength-switching elements. These hybrid designs enable optimization of key tradeoffs, such as achieving both fast actuation and low insertion loss. The main trade-off is increased design and fabrication complexity, as many hybrid solutions are still emerging and require specialized manufacturing processes.

Following is a table summarizing the OCS Features & Technology capabilities.

Features vs Technology	Robotic	Beam Steering MEMS/Liquid Crystal/Piezo Electric	Silicon Photonics MZI/MEMS	Metasurface
Radix	Large	Med	Low-Med	Large
Insertion Loss (dB) *	Low	Med	High	Med
Dark Fiber Switch **	Yes	Some	Yes	Yes
Switch/Reconfiguration Times	Slow (Sec-Min)	Med (msec)	Fast (nsec to usec)	Fast (usec)
Simultaneous Switch/Reconfiguration connections per Switch	No	Yes	Yes	Yes
Integrated OPM per Port	Possible	Some	Some	Some
Integrated SOA per Port	Possible	Possible	Some	Possible
Integrated VOA per Port	Possible	Some	Some	Some
Latched switch w/o Electrical Power	Yes	No	Some***	No

* Insertion Loss typically increases as the # of Radix Increases. Insertion Loss can be reduced with an integrated SOA.

** No light is required at the Optical Input Port for the switch to perform the optical connection to another Optical Port

*** Silicon Photonics with Phase Change Material (PCM)

Depending on system requirements—such as resilience, speed, or operational independence—users can choose or combine OCS technologies to meet specific performance needs.

5. OCS Applications for AI Datacenters

Current OCS deployment and use cases:

A. OCS use case in EPS Spine Replacement

In this scenario, the spine-layer electrical packet switches (EPS) of a data center (DC) are replaced with Optical Circuit Switching (OCS). In current deployments, this function is typically implemented using a large-radix OCS system with slow, infrequent reconfiguration cycles. When the spine layer is replaced, all leaf EPS devices become directly connected—from a packet-forwarding perspective—through a flat OCS layer.

The primary benefit of OCS spine replacement is the dramatic reduction (or complete elimination) of power consumption associated with spine-layer EPS switches, which is significant in hyperscale data centers. For example, if we assume a large DC uses sixty-four 16-slot chassis-based spine switches, each consuming approximately 30 kW (assuming 400G with half-populated coherent optics), the spine layer alone would draw roughly 1.9 MW at maximum load. In another example, Google reported in their Jupiter Evolving paper [1] that replacing EPS with OCS at the spine layer yielded a 41% reduction in network power consumption.

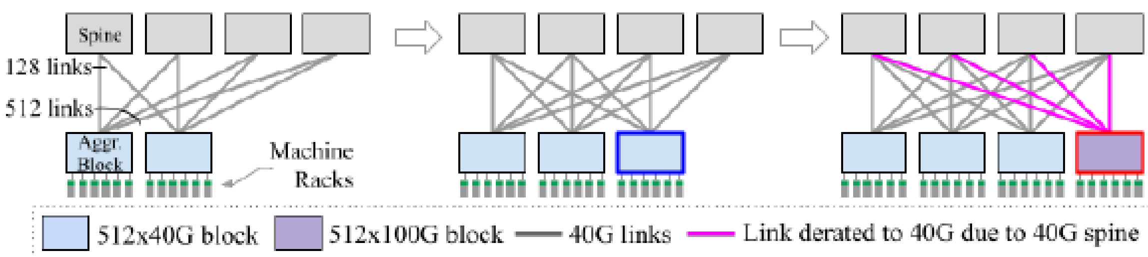
A secondary benefit of spine-layer OCS replacement is its transparency to, and compatibility with, leaf-switch and interconnect upgrades. Because the optical spine does not impose speed constraints, operators no longer need to upgrade the spine switches to match the higher rates of evolving leaf switches in order to remove bandwidth bottlenecks.

While spine replacement offers substantial benefits, it also presents challenges. To optimize OCS usage while minimizing capacity overprovisioning, traffic flows must be predictable or continuously monitored, conditions that are not always guaranteed in dynamic data-center environments.

Nevertheless, OCS spine replacement provides several additional advantages noted by Google in Jupiter Evolving [1], including a 30% reduction in CAPEX, a 5× increase in network throughput, and faster deployment timelines. As a result, operators must weigh these benefits against risks, architectural complexities, and operational maturity.

We expect that future improvements in both OCS radix and switching times will further reduce barriers to adopting OCS-based spine replacement on a large scale.

The following figures (Figure 1 and 2) are taken from the “Jupiter Evolving” paper [1]. They illustrate Google architecture deployment before and after using OCS, replacing the EPS Spine Switch with OCS Switch.



<u>Figure 1: A 3-tier clos network comprising machine racks, their aggregation blocks, and 40Gbs spine blocks connecting the aggregation blocks. All spines are deployed on Day 1. Blocks deployed on Day 2 are outlined in blue while those deployed on Day N are highlighted in red. Links from 100Gbps aggregation block are derated to 40Gbps due to the 40Gbps spine.</u>

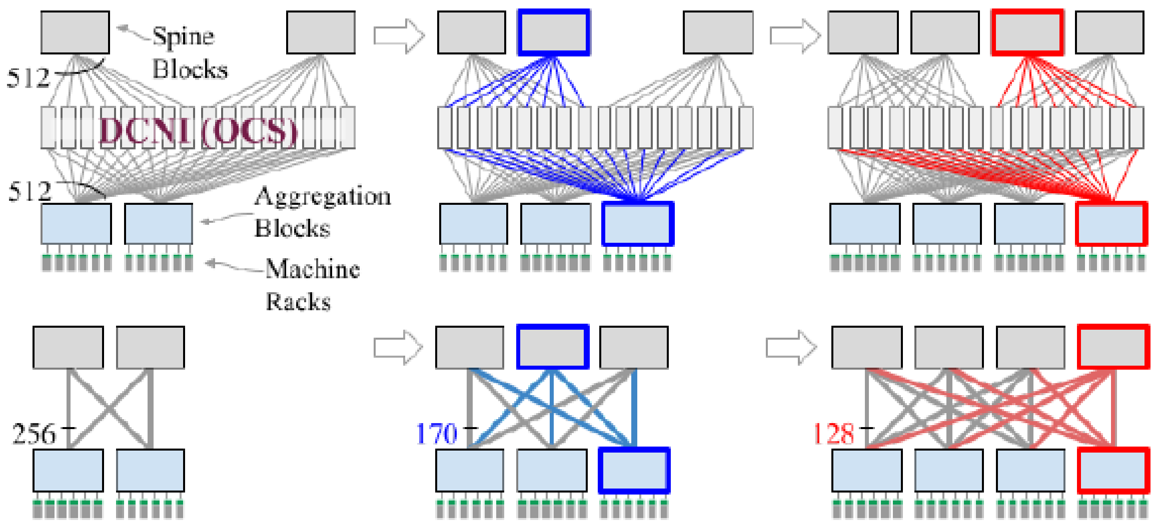


Figure 2: An optical switching layer, DCNI (top), enables incremental expansion in Jupiter while achieving full burst bandwidth among the aggregation blocks (logical topology at the bottom). The DCNI layer is implemented using OCS and allows incremental rewiring of the fabric as new blocks are added.

The above writing is a high-level summary and understanding of the *Jupiter Evolving: Transforming Google's Datacenter Network via Optical Circuit Switches and Software-Defined Networking* paper [1]. Readers are encouraged to consult the reference for more details and conclusions.

B. OCS use case in Static to Dynamic AI Cluster Reconfiguration

In Google's TPUv4 (and later) clusters, OCS systems are deployed between TPU servers to create a physically reconfigurable AI cluster. This architecture enables sub-dividable compute topologies and provides mechanisms to reroute traffic around failures. In the TPU design, servers are arranged into hardwired blocks known as Cubes. Each Cube contains 16 servers, and each server hosts 4 TPUs, for a total of 64 TPUs per Cube. In TPUv4, 64 of these Cubes form a Superpod, resulting in a total of 4096 TPUs per Superpod.

Within a TPUv4 Superpod, each Cube connects to three groups of sixteen OCS devices—48 OCS units in total. All OCS devices in the TPUv4 system are 128-port MEMS-based optical circuit switches. These switches can be dynamically reconfigured on a per-job basis through Google's sophisticated software control plane.

The OCS fabric in TPU Superpods provides several key benefits:

1. **Topology Reconfiguration for Performance:** OCS enables Cubes to be interconnected in different topologies on a per-job basis. At the Superpod level, the reconfigurable optical fabric makes it possible to create job-specific network configurations—such as torus topologies—based on the application's communication patterns. Matching the network topology to job parameters improves both performance and power efficiency. For example, training large language models on TPUv4 Superpods achieved a 3.3× improvement in performance compared to static configurations and a 9% reduction in power consumption relative to a Superpod using an EPS-based DCN fabric [11].
2. **Improved System Availability and Failure Bypass:** The OCS interconnect increases system availability. In the event of failure, OCS can bypass failed Cubes. This is crucial because TPU traffic can traverse servers inside a Cube. For example, with 99.9% server availability, a static configuration supports a 1024-TPU slice size with only 25% effective throughput, whereas the reconfigurable Superpod supports the same slice size with 75% effective throughput [2].
3. **Cluster Sub-division and Isolation:** Cubes—or groups of Cubes—can be logically isolated from the rest of the Superpod. This allows multiple independent jobs to run simultaneously in an isolated fashion, improving Superpod utilization as well as security.

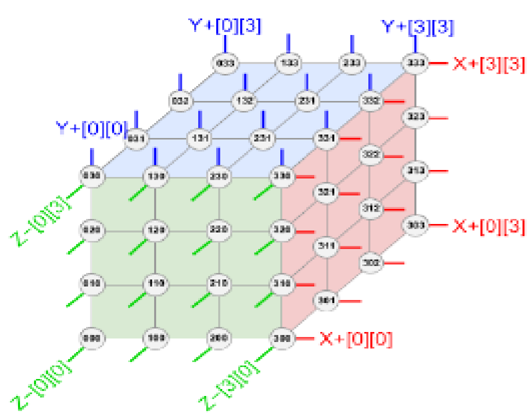
Additional Architectural Notes: OCS connectivity exists between Cubes, not within a Cube. Inside each Cube, interconnections are electrical rather than optical. Additionally, because of the Cube's internal topology, not all TPUs have direct access to the OCS fabric; "non-surface" TPUs must forward traffic through other TPUs to reach the optical layer.

As noted earlier, a significant portion of the TPU architecture’s complexity resides in the control plane, which manages Cube assignment for each job, addressing, routing, and failover behavior. Newer TPU generations scale this architecture further by incorporating larger OCS devices, increasing the number of optical switches, and adopting faster Inter-Chip Interconnect (ICI) speeds. Despite these enhancements, the fundamental role of the OCS fabric remains the same.

Figures and References: Figures 3, 4, 5, and 6 referenced here are taken from the following sources:

- TPU v4: An Optically Reconfigurable Supercomputer for Machine Learning with Hardware Support for Embeddings [2]
- Resiliency at Scale: Managing Google’s TPUv4 Machine Learning Supercomputer [3]
- How to Scale Your Model (A Systems View of LLMs on TPUs) [10]

These figures illustrate the overall Cube design—including servers, TPUs, and both inter-Cube and intra-Cube connectivity. While the diagrams show OCS connectivity on six “faces” of the Cube, the system is organized into three groups of OCSs, corresponding to the X, Y, and Z dimensions of the fabric.



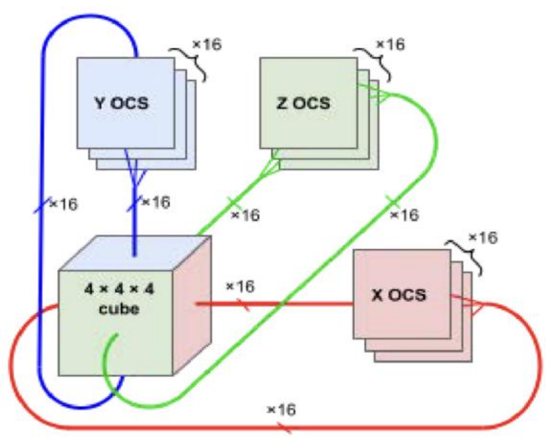


Figure 3 (left): A hardwired TPU cube (4x4x4).

Figure 4 (right): Connectivity of a 4x4x4 cube to 3 groups of OCSs.

A single Cube with 64x TPUv4 resides in a single rack as shown in Figure 5.

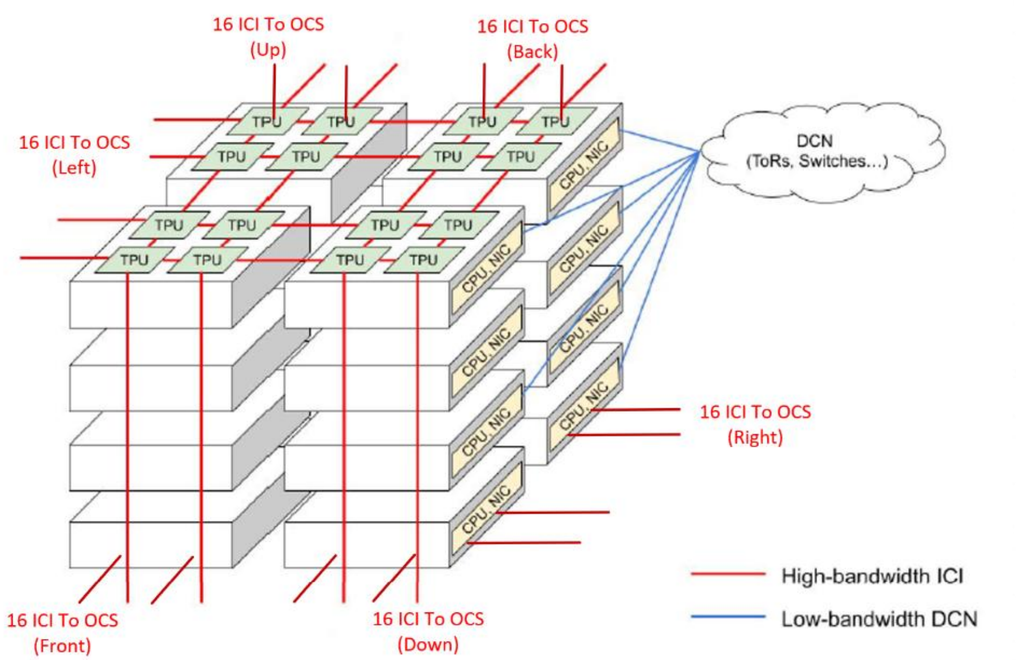


Figure 5: Rack view of the 4x4x4 TPU cube.

The following diagram shown in Figure 6 illustrates how Cubes can be rearranged into smaller units using X, Y, Z OCS, as needed for a particular job. Within a Superpod sub-unit, the connectivity can be matched to a specific job training.

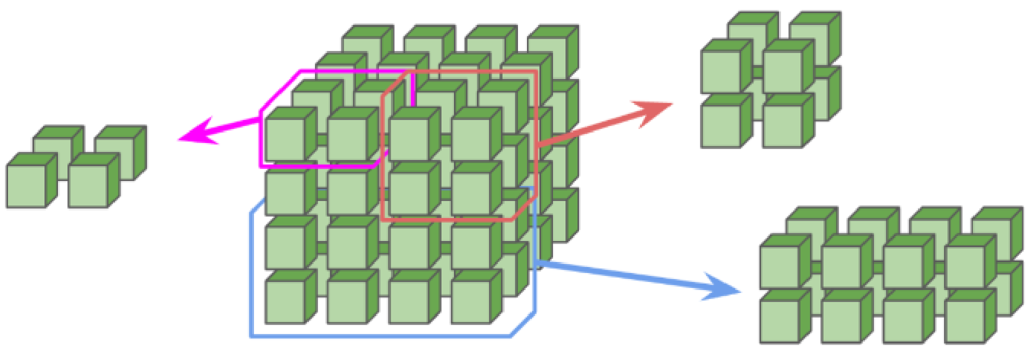


Figure 6: Reconfigurable cubes in a TPU superpod.

The above writing is the high-level summary and understanding of the following research papers [2], [3] and [10]. Readers are encouraged to read these references for further details and conclusions.

C. OCS use case in Hardware Failure Resiliency

As AI and high-performance computing systems continue to scale, hardware failures become a normal operating condition rather than an exception. Switches, links, and optical modules fail regularly, and even short outages can disrupt long-running workloads, reduce system utilization, and increase operational costs. Traditional network recovery mechanisms rely on software-based rerouting, which often cannot fully restore performance and can fail entirely when physical-layer faults disconnect servers or entire racks.

Optical Circuit Switching (OCS) introduces a new approach to resiliency by making the physical network itself programmable. Instead of treating cable and optical paths as static infrastructure, OCS enables dynamic reconfiguration of physical network connections in response to hardware failures.

In an OCS-enabled architecture, optical switches are incorporated into the network fabric and controlled by a software orchestration system. When a failure is detected, the control plane automatically reconfigures optical connections to bypass the faulty component and reroute traffic through available spare pathways. This physical-layer response enables recovery from failures that would otherwise cause application downtime.

A key advantage of this model is rapid restoration of full performance. Unlike traditional rerouting techniques— which often leave the network in a degraded or bandwidth-constrained state—optical reconfiguration restores the intended connectivity and available bandwidth. For AI and distributed computing workloads, this means jobs can continue running without restarts, checkpoint rollbacks, or prolonged slowdowns.

OCS-based resiliency also reduces the need for costly over-provisioning. Instead of duplicating every network path or maintaining a large amount of idle backup infrastructure, operators can maintain a shared pool of spare optical paths that protect many components simultaneously. This improves availability while reducing capital costs, power consumption, and operational overhead.

By enabling automated, physical-layer recovery, optical circuit switching transforms the network from a passive transport layer into an active resiliency platform. The result is higher system uptime, more predictable performance, and improved efficiency for large-scale AI and data-center environments.

Figure 7 (from NVIDIA's recent publication [4]) illustrates how OCS can be used to provide protection and bypass capabilities in the event of hardware failures in the EPS spine and leaf layers.

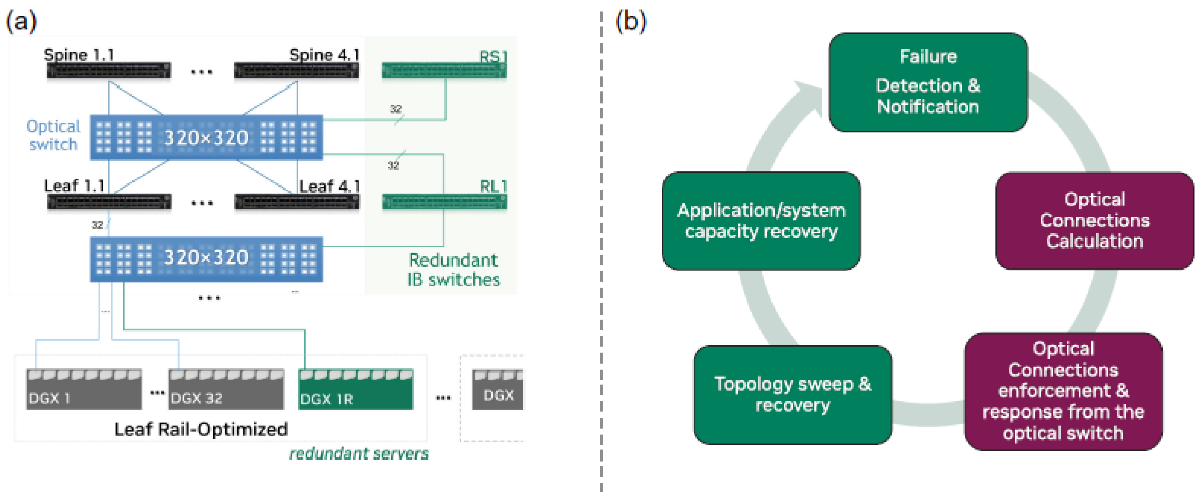


Figure 7: (a) Optically enabled switch resilience via the addition of OCS and redundant electrical switches. (b) control-plane steps for recovery in case of switch failures.

The above writing is the high-level summary and understanding of the following research paper [4]. Readers are encouraged to read this reference for further details and conclusions.

Possible future OCS deployment and use cases:

D. OCS use case in Dynamic In-Workload AI Cluster Reconfiguration for AI Cluster performance optimization.

One emerging future use case for dynamic AI-cluster reconfiguration using OCS is highlighted in an academic research effort called ACTINA (Adaptive Circuit Switching Techniques for AI Networking Architecture), authored by Columbia University and NVIDIA Corporation [5]. The

ACTINA paper identifies the core limitations of today's AI networking architectures and proposes a dynamic circuit-switching–based approach designed specifically for next-generation AI data centers.

Current AI infrastructure faces rapidly escalating power demands—projected to increase by as much as 200× by the end of the decade—levels that traditional EPS-based topologies cannot sustainably accommodate. Static EPS designs also lack the flexibility needed to support the highly dynamic and irregular communication patterns of modern AI workloads, making them difficult to optimize and slow to recover from failures. Meanwhile, existing OCS deployments—such as Google's OCS Fat-Tree 2-Layer (OCSFT-2L) architecture for EPS spine-layer replacement and the OCS-enabled TPUPod 3D Torus—improve efficiency but still suffer from relatively slow reconfiguration, limited adaptability, and bandwidth under-utilization during active training workloads. These constraints ultimately reduce achievable training performance at scale.

To address these gaps, ACTINA introduces OCS-BCube, an optical-circuit-switched topology inspired by the multi-dimensional connectivity principles of the original BCube architecture but redesigned entirely for GPU-centric AI training clusters rather than modular container data centers. Unlike classic BCube—where servers act as both end hosts and intermediate forwarding elements—OCS-BCube retains the multi-dimensional design concepts but removes all electrical packet forwarding. Instead, it employs a custom silicon-photonics multi-port reconfigurable transceiver, integrated directly into each GPU, to provide direct all-to-all connectivity along each BCube dimension through OCS.

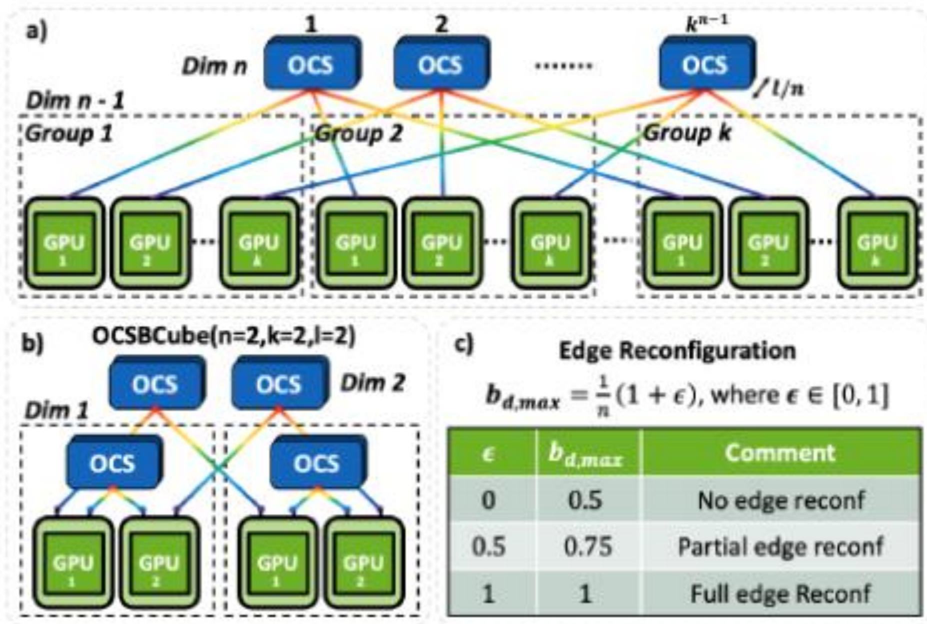
This Silicon Photonic (SiP) module is built using Mach-Zehnder Interferometer (MZI) components and a DWDM comb-laser. It can dynamically reassign wavelengths on a per-port basis, enabling real-time bandwidth steering and flexible all-to-all GPU communication across data, pipeline, and tensor parallelism domains [6]. With ACTINA's traffic-aware control plane, OCS-BCube supports fine-grained, intra-iteration optical reconfiguration, improving bandwidth utilization, reducing contention hot spots, and enabling more efficient scaling for large AI workloads.

The ACTINA evaluation shows that OCS-BCube can deliver:

- **Up to 1.84× faster iteration times** compared to an OCS-enabled 3D Torus, while maintaining similar energy efficiency.
- **Equivalent iteration times to OCSFT-2L**, but with up to **1.72× lower energy consumption**.
- **Up to 1.75× better tokens per joule**, due to dynamic, in-workload reconfiguration and edge bandwidth steering.

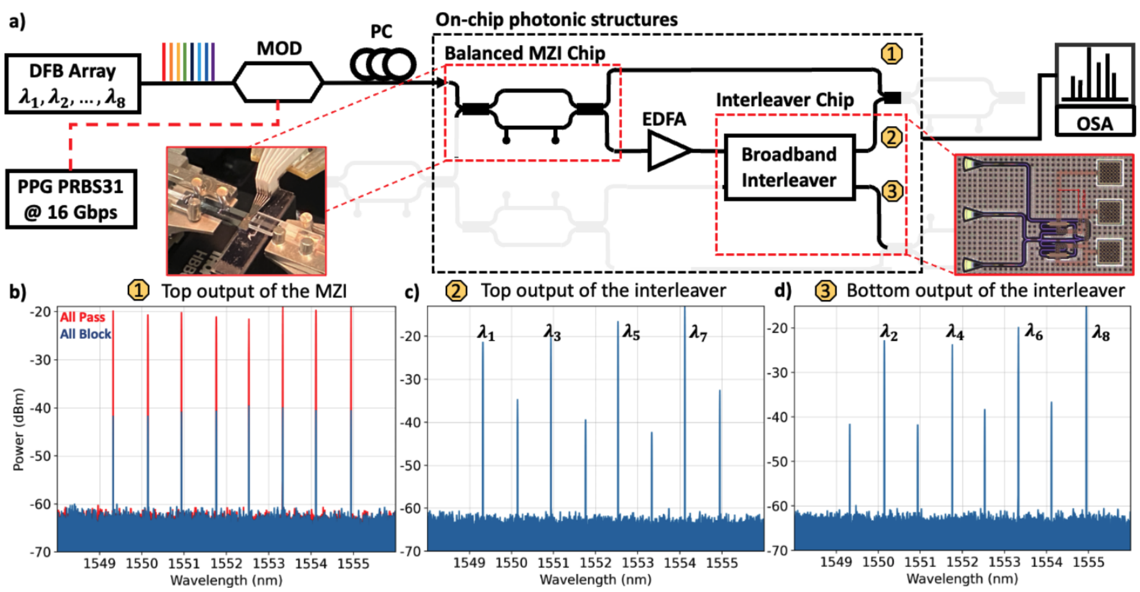
These gains are enabled by the combined use of (1) the OCS-BCube topology and (2) the new architectural concept of a multi-port, wavelength-reconfigurable transceiver integrated directly into each GPU—something not feasible in traditional EPS or earlier OCS designs.

Figure 8, taken from the ACTINA paper [5], illustrates the design and key characteristics of the proposed OCS-BCube topology and its edge-level optical reconfigurability.



<u>Figure 8: (a) OCSBCube topology with n dimensions and k ports per OCS. (b) Example of OCSBCube with n=2, k=2, l=2. © Maximum domain bandwidth in a 2-dimensional OCSBCube for different degrees of edge reconfigurability.</u>

The following Figure 9 is taken from an OFC short paper [6]. It shows the experiment setup for the next generation Multi-Port and Wavelength Reconfigurable Transceiver concept that can be used in the new OCSBCube Topology concept.



<u>Figure 9: (a) Schematic of experimental setup: 8 evenly spaced wavelengths generated by Distributed Feedback (DFB) lasers modulated at 16 Gbps and traverse the upper section of the structure, featuring a MZI cascaded with a broadband interleaver. Optical spectra (b,c,d) depict the focused signals in the target wavelength range for various routing operations.</u>

The above writing is the high-level summary and understanding of the following research papers [5] and [6]. Readers are encouraged to read these references for further details and conclusions.

E. OCS use case in Reconfigurable GPU's Interconnect for Optimizing Mix of Experts (MoE) Training

OCS-based network topologies are emerging as promising architectures for enabling more efficient bandwidth allocation in machine-learning (ML) training, particularly for workloads with highly structured and localized communication. Among the various parallelization strategies used in distributed ML, Expert Parallelism (EP)—central to Mixture-of-Experts (MoE) models—is characterized by nondeterministic, all-to-all communication across a distributed group of "experts" residing on different GPUs. The set of communicating GPUs and their traffic volumes vary from iteration to iteration and can only be determined *after* the computation phase and *before* all-to-all communication steps. This makes it difficult to reconfigure an OCS fabric in advance without stalling the training pipeline.

At the same time, EP has the highest communication volume among major parallelism strategies, including data parallelism (DP) and pipeline parallelism (PP). As a result, supporting EP with OCS can meaningfully improve overall training efficiency.

Researchers have addressed this challenge with a hybrid electrical/optical interconnect architecture called MixNet [7], which pairs a reconfigurable optical fabric with a specialized runtime system designed to exploit both temporal and spatial locality in EP traffic. MixNet provides GPUs across different servers with dynamic, reconfigurable bandwidth that exceeds what the global packet-switched network can deliver (as shown in Figures 10 and 11). EP communication flows exclusively over this optical fabric, while DP and PP traffic continue to use the global EPS packet network.

MixNet is based on two key insights:

1. **EP communication is localized rather than global.**
   In practice, only a small group of GPUs (typically 8–32) participate in all-to-all communication during an iteration, and these groups are isolated from the rest of the system. This allows these GPUs to be connected through small-radix OCS devices with millisecond-scale reconfiguration times. Importantly, the OCS can be reconfigured during the computation phase, ensuring that EP traffic is uninterrupted when communication begins.
2. **EP traffic patterns change gradually between iterations.**
   Although communication groups vary per iteration, consecutive iterations differ only slightly. Therefore, the OCS fabric can be reconfigured according to a predicted traffic pattern derived from the previous iteration, and this prediction remains sufficiently accurate for allocating bandwidth without degrading training progress.

By combining these two insights, MixNet dynamically reconfigures the OCS fabric *during* the computation portion of an iteration, so the correct optical links and bandwidth allocations are ready by the time the EP all-to-all phase begins.

MixNet represents one of the first community efforts to support dynamic, in-workload OCS reconfiguration for MoE training. Simulation results from the MixNet paper show that this approach reduces training time for the Mistral 8×7B MoE model by 1.6× compared to a static fat-tree packet network.

Figures and Architectural Overview Figures 10 and 11 (from the MixNet paper [7]) illustrate the MixNet network architecture and its optimizations for distributed MoE training. The diagrams highlight the integration of three types of interconnect fabrics:

1. **Local Scale-Up Fabric (NVSwitch)**
   Connects GPUs within the same server, providing high-bandwidth, low-latency communication for tensor parallelism (TP) and intra-server data transfers.

2. **Global Scale-Out Fabric (EPS)**
   Uses Ethernet or InfiniBand to connect servers across the cluster, supporting DP and PP traffic patterns.
3. **Regional OCS Fabric**
   Handles the dynamic and non-uniform all-to-all communication patterns characteristic of EP. MixNet divides the cluster into regional domains, each equipped with a reconfigurable OCS to service localized EP communication efficiently.

This hybrid design delivers scalability, adaptability, and improved cost efficiency by leveraging the strong locality inherent in MoE traffic patterns and aligning physical interconnect resources with workload demands.

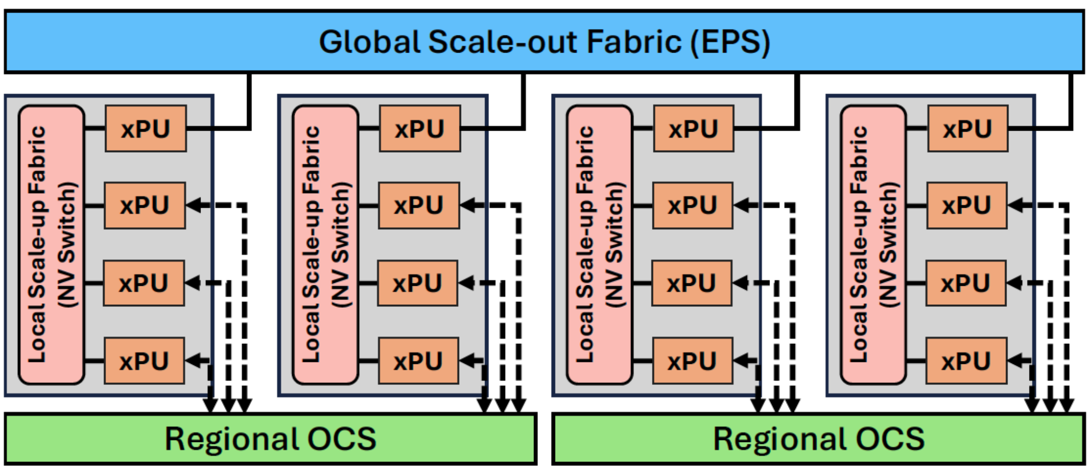


Figure 10: The MixNet network architecture.

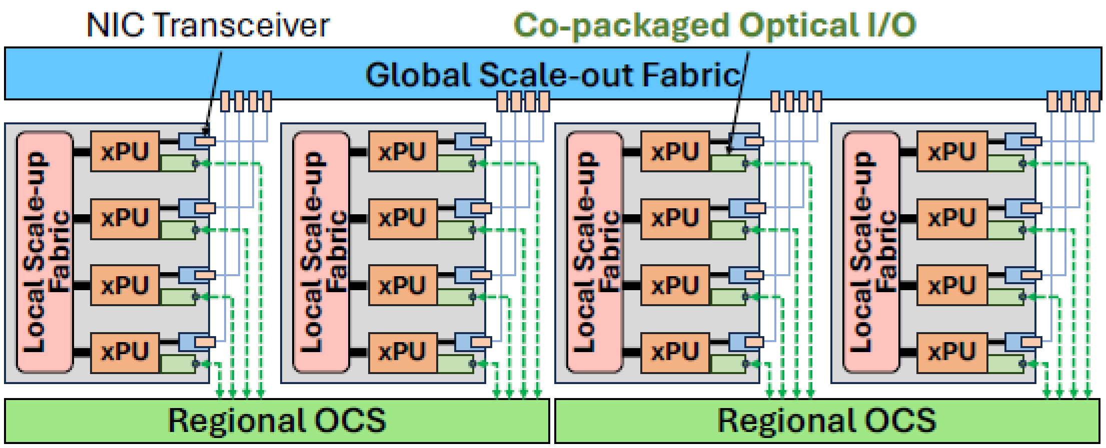


Figure 11: MixNet with co-packaged optical ports directly attached to xPUs.

The above writing is the high-level summary and understanding of the following research paper [7]. Readers are encouraged to read this reference for further details and conclusions.

6. Potential Future OCS Use Cases for other applications

OCS in quantum networks

The quantum networking landscape faces a fundamental bottleneck: the extreme fragility of quantum information. In traditional EPS networks, signals undergo multiple optical-electrical-optical (O-E-O) conversions. While this is acceptable for classical bits, it is fatal for quantum bits (qubits). According to the No-Cloning Theorem, quantum states cannot be copied or measured without being destroyed. An EPS switch must “read” a packet in order to route it, which effectively measures the photon. This measurement causes immediate decoherence and collapses the quantum state, making standard buffered switches physically incapable of distributing entanglement or supporting a future “quantum internet.”
OCS systems address this limitation by providing a transparent, all-optical switching path. Because OCS devices redirect light without converting photons into electrical signals, the

quantum states remain entirely within the optical domain. This transparency allows entangled photons to pass through the switch without collapsing their quantum state, preserving the high fidelity required for quantum teleportation, entanglement distribution, and distributed quantum computing [8]. In essence, an OCS functions more like a clear glass window than a processing device—it simply guides the photons while maintaining their delicate phase, polarization, and other quantum properties.

Achieving predictable latency variance with OCS

Latency-fairness in AI/HPC. Large-scale AI and HPC applications are extremely sensitive to network jitter, as bulk-synchronous operations such as barriers and AllReduce can be stalled by even a single delayed packet [9]. Traditional packet-switched networks naturally introduce stochastic latency variation due to queuing, buffer contention, and congestion dynamics, producing “stragglers” that slow down collective operations and degrade overall performance. Optical Circuit Switching (OCS) mitigates this by creating dedicated, physical-layer lightpaths that avoid electrical store-and-forward processing altogether. With deterministic, near-zero-jitter paths that are fully isolated from cross-traffic, OCS ensures that synchronization signals propagate with uniform and predictable latency. This physical-layer determinism eliminates the long-tail delays common in shared packet fabrics, sustaining high application performance across large-scale clusters.

OCS-enabled data center resource disaggregation.

Resource disaggregation today relies on packet-switched networks to connect servers with pooled resources—such as accelerators, memory, and storage—while providing high-bandwidth, low-latency interconnects [12]. However, this approach faces two major challenges. First, the port count of packet switches decreases as per-port speeds increase, which forces traffic to traverse more hops in a hierarchical packet-switched network, resulting in higher end-to-end latency. Second, the effective reach of high-speed copper links continues to shrink at higher signaling rates. To extend reach for inter-rack connectivity, fiber optics combined with O-E-O conversion must be introduced, but this further adds latency, power consumption, and cost. OCS technology offers compelling advantages for this use case.

Because OCS is inherently rate-agnostic, it allows bandwidth to scale efficiently—such as through DWDM—without reducing port count. Additionally, OCS avoids O-E-O conversion delays and eliminates queueing-related latency, enabling significantly lower end-to-end latency for resource disaggregation architectures.

7. Conclusion

As of 2026, OCS has evolved from a specialized technology into a cornerstone of hyperscale data-center networking infrastructure, driven primarily by Google's successful large-scale deployment of its Jupiter fabric and TPU clusters. By eliminating power-intensive O-E-O conversions, OCS has enabled substantial reductions in power consumption and capex, while delivering the ultra-low-latency and jitter-free performance required by modern workloads.

While Google currently leads in production-scale deployment, it is widely expected that other industry players are actively exploring or developing OCS-enabled architectures. This growing industry momentum highlights the significant potential of optical switching technologies and signals rapid market expansion as the data-center ecosystem shifts toward more optically enabled infrastructure designs.

8. Table of Glossary

Term	Meaning
Capex	Capital expenditure
DC	Data center
DP	Data Parallelism
DWDM	Dense Wavelength Division Multiplexing
EPS	Electrical packet switching
EP	Expert Parallelism
ICI	Google's Inter-Chip Interconnect
LCoS	Liquid Crystal on Silicon
MEMS	Micro-Electro-Mechanical Systems
MoE	Mixture of Experts
MZI	Mach-Zehnder Interferometer
OCP	Open Compute Project
OCS	Optical circuit switching
O-E-O conversion	Optical-electrical-optical conversion
Opex	Operational expenditure
OPM	Optical Power Monitoring

PP	Pipeline Parallelism
SOA	Semiconductor Optical Amplifier
SiP	Silicon Photonics
Switch radix	Port count on a switch
TPU	Google’s Tensor Processing Unit
VOA	Variable Optical Attenuator
WSS	Wavelength Selective Switches

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